

PROPOSAL: ASIAN ROVER CHALLENGE (AsRC) 2026

1. Executive Summary

The Asian Rover Challenge (AsRC) is a premier student-led Mars robotics competition designed to prepare the next generation of engineers for the challenges of planetary habitation and infrastructure development. Hosted in Kerala, India, and organized by RISA (Robotic Systems Integrators Association) in collaboration with Mar Athanasius College of Engineering (MACE), AsRC brings together engineering students from across Asia to design, build, and operate advanced Mars-capable robotic systems in realistic mission environments.

The Paradigm Shift: Beyond Initial Exploration

Unlike traditional rover competitions that focus primarily on initial exploration and scientific discovery, AsRC is built around a fundamentally different question: **What happens after exploration?**

For radiography and decades, prestigious international competitions such as the University Rover Challenge (URC) in the USA, the European Rover Challenge (ERC) in Poland, the Canadian International Rover Challenge (CIRC), and the Anatolian Rover Challenge have established excellent global standards for student planetary robotics. Their missions successfully replicate humanity's first encounters with a planetary surface—exploring unknown terrain, collecting geological samples, identifying scientific targets, and demonstrating autonomous mobility.

The Asian Rover Challenge respects and builds upon this foundation but shifts the paradigm entirely. AsRC is set in a near-future Mars, approximately in the 2040s, where the era of initial exploration has already been completed. High-resolution maps exist, sample caches have been deposited, and geological surveys are complete. The challenge facing humanity is no longer discovering Mars—it is learning how to live, operate, and build there.

In this context, robotic systems must evolve beyond being mobile scientific instruments. They must become infrastructure-enabling machines—"mobile workers"—capable of supporting long-term human presence. The competition, therefore, introduces a new generation of mission scenarios focused on maintenance, recovery, logistics, deployment, and construction.

Democratizing Space Robotics for Asia

While elite competitions in the United States and Europe represent the pinnacle of student aerospace engineering, participating in them remains an inaccessible dream for the vast majority of talented student teams across India and Asia. Prohibitive travel expenses, complex international shipping logistics for heavy rover systems, and stringent visa barriers prevent hundreds of brilliant young minds from showcasing their innovations on a global stage.

By hosting AsRC at MACE in Kerala, we establish a world-class, financially viable, and logistically accessible alternative. This competition will democratize elite space robotics in the

region, offering Indian and Asian teams a high-caliber arena to test their systems without the crippling financial burdens of Western-centric travel.

2. Objectives

The Asian Rover Challenge (AsRC) is designed to serve as a comprehensive platform for advancing robotics, space technology, and engineering education while preparing students to tackle the challenges of future planetary habitation.

- **Promote Robotics and Space Technology Education:** Increase awareness and interest in robotics, automation, autonomous systems, and planetary exploration among students across Asia. By exposing participants to real-world engineering challenges, the competition encourages students to explore emerging technologies and apply theoretical concepts in meaningful engineering applications.
- **Fostering Indian and Asian Talent (Democratization):** Provide an elite, highly accessible, and affordable platform for teams from SAARC and Asian nations, eliminating the massive logistical and financial barriers associated with traveling to Western-based competitions.
- **Establish MACE's Global Academic Repute:** Position Mar Athanasius College of Engineering as a premier pioneer in space-tech education. Hosting an international event of this scale will draw global media attention, research collaborations, and enhance institutional standing in international academic circles and national accreditations (such as NBA and NAAC).
- **Encourage Multidisciplinary Collaboration:** Modern planetary missions require expertise from multiple engineering domains working together. The competition encourages collaboration among students from mechanical, electronics, electrical, computer science, artificial intelligence, mechatronics, and civil engineering.
- **Advance Infrastructure-Oriented Planetary Robotics:** Lead the academic transition from exploration-based rovers to infrastructure-oriented planetary robotics. Through mission requirements involving autonomous navigation, robotic manipulation, maintenance operations, and construction tasks, participants are motivated to develop innovative solutions for sustaining extraterrestrial settlements.
- **Develop Future Technology Leaders:** Cultivate leadership, project management, budgeting, and systems-engineering skills among participants, helping to train the next generation of space and automation industry professionals.

3. Background and Need

The twenty-first century has witnessed a historic surge in global interest in space exploration. Space agencies like ISRO, NASA, and ESA, alongside private aerospace corporations, are actively developing technologies to support permanent human presence beyond Earth. Planetary rovers serve as the critical vanguard of these missions, functioning as the primary tools for construction, logistics, and maintenance in environments hazardous to humans.

However, a major gap exists in modern engineering education. While students gain strong theoretical knowledge through coursework, they rarely have opportunities to apply concepts

from multiple disciplines in a highly integrated, high-pressure environment. Rover competitions have emerged worldwide as one of the most effective methods for addressing this challenge, offering unparalleled experiential learning.

Currently, existing international rover events are concentrated in the Western hemisphere, creating a geographical and economic barrier for Asian universities. Furthermore, these existing competitions focus almost exclusively on the early exploration phase. As humanity prepares to establish semi-permanent bases on the Moon and Mars, there is an urgent industrial and academic need for engineering programs to pivot toward maintenance, logistics, and infrastructure support.

The Asian Rover Challenge directly addresses this dual need. It provides a highly challenging, multidisciplinary engineering experience tailored specifically to the next generation of space habitation challenges, hosted in a geographically and financially accessible location for the eastern hemisphere.

4. Why MACE?

Mar Athanasius College of Engineering (MACE), Kothamangalam, with its legacy spanning more than six decades of academic excellence, provides the perfect ecosystem for hosting an international technical event of this scale.

Strong Robotics and Innovation Ecosystem (Team Srishti)

At the center of MACE's technical culture is Team Srishti, the college's flagship student rover team. Having competed firsthand in prestigious international planetary rover competitions, Team Srishti possesses deep technical and operational knowledge of rover design, systems integration, and competition dynamics. Their practical experience ensures that MACE has the direct expertise required to design, construct, and judge a world-class simulated Martian arena.

Advanced Laboratories & Prototyping Facilities

MACE's newly established Robotics and Automation Laboratory, combined with its state-of-the-art FabLab, offers participants and organizers access to advanced prototyping, digital fabrication, and electronics testing equipment. Extensive engineering departments (Mechanical, Civil, Electrical, Electronics, and Computer Science) provide a robust network of laboratories and machine shops to support participating teams with emergency repairs and technical troubleshooting during the event.

Strategic Location and Accessibility

Situated in Kothamangalam, the "Gateway to the High Ranges," the campus offers the expansive physical space required to build a realistic, large-scale simulated Martian terrain. Simultaneously, the college remains highly accessible:

- **Air Connectivity:** Cochin International Airport (COK) is located just one hour away from

the campus, providing direct flights for national and international teams.

- **Rail and Road Connectivity:** Major railway hubs like Aluva and Ernakulam connect the campus directly to all parts of India.

Collaboration with Mar Athanasius College of Arts and Science (MACAS)

By collaborating with the sister institution, MACAS, MACE can utilize expanded infrastructure, including additional auditoriums, seminar halls, extensive dining systems, and accommodation spaces, ensuring a seamless experience for hundreds of visiting competitors and delegates.

5. Proposed Competition Structure

The Asian Rover Challenge is organized as a rigorous, two-stage competition to evaluate both theoretical engineering design and physical operational excellence.

Stage 1: Online Qualification Round

Teams submit a comprehensive engineering package, which is evaluated by a panel of industry experts:

- **System Design Report & Video Presentation**
- **Science Report & Cost Analysis**
- **Settlement Design Report & 3D CAD Model** (Focusing on a proposed Martian settlement layout and rover docking station).

Stage 2: On-Site Finals (3-Day Field Competition)

The top-ranked qualifying teams bring their rovers to the custom-built simulated Martian terrain at MACE to execute high-stakes physical missions:

- **Mission 1: Maintenance & Repair Operation:** Restoring power and communications to a dormant infrastructure asset by interacting with control panels.
- **Mission 2: Sample Recovery & Transport:** Safely retrieving and transporting cached resource payloads across difficult terrain.
- **Mission 3: Pavement Construction Task:** Actively deploying interlocking tiles to construct a traversable path over soft regolith.
- **Mission 4: Science Investigation Task:** Conducting an in-situ scientific study of a designated geological anomaly.

6. Expected Participation

The inaugural edition of the Asian Rover Challenge is planned as a high-footfall regional event with an expanding international footprint:

Category	Expected Scale / Footfall
Competing Rover Teams	25 – 30 elite teams (India, SAARC, and Southeast Asia)

Direct Student Competitors	500 – 900+ multidisciplinary engineers
Faculty Advisors & Delegates	60+ academic representatives
Industry Experts, Judges & Speakers	30+ robotics and space-tech professionals
On-Campus General Footfall	1,500+ (Students, tech enthusiasts, and media)

7. Infrastructure Requirements

To execute this event professionally, MACE will dedicate and develop the following infrastructure:

1. **Simulated Martian Arena:** A dedicated outdoor area modeled with rocky terrain, loose regolith fields, steep slopes, and obstacles representing the Martian surface.
2. **Mission Control & Team Pits:** Covered, secure workspaces equipped with reliable power, internet connectivity, and workbenches for rover calibration and battery charging.
3. **Technical Presentation & Seminar Halls:** Air-conditioned auditoriums for team registrations, safety briefings, expert guest lectures, and the final awards ceremony.
4. **Logistical Support:** Structured accommodation, dining plans, and first-aid medical facilities to support visiting students and international delegates over the course of the event.

8. Budget Framework

To successfully execute an event of this magnitude while maintaining strict fiscal responsibility, we have optimized the budget to ensure maximum institutional impact for every rupee spent. The estimated financial requirements are detailed below:

No.	Expenditure Item	Description	Estimated Cost (₹)
1.	Simulated Martian Terrain / Arena	Construction of realistic Martian landscaping, obstacle zones, and physical mission assets (control panels, sample caches, interlocking pavement tiles).	₹1,00,000
2.	Judges' Stay &	High-quality	₹40,000 (Est.)

	Accommodation	boarding, lodging, and local transit for invited space scientists, industry professionals, and technical evaluators.	
3.	Social Evening Event	Networking dinner and cultural evening to connect visiting international teams, local students, faculty, and industry sponsors.	₹30,000
4.	Trophies, Medals & Awards	High-quality mementos, custom certificates, and cash prizes for winning teams across individual mission categories.	₹20,000
5.	Refreshments (5-Day Event)	Snacks, water, and working lunch provisions for judges, volunteers, event staff, and organizers over the 5-day on-site duration.	₹10,000
Total	Estimated Budget Requirement		₹2,00,000

Note: Revenue offsets will be actively pursued through team registration fees, corporate sponsorships, and industry partnerships to minimize direct institutional expenditure.

9. Expected Outcomes & Institutional Gains for MACE

Hosting the Asian Rover Challenge yields immediate, high-value returns for Mar Athanasius

College of Engineering:

- **Global Academic Visibility:** Establishing AsRC puts MACE on the international aerospace map, significantly boosting institutional prestige, international student exchange potential, and scores for key national ranking frameworks (NIRF, ARIIA) and accreditation bodies (NBA, NAAC).
- **Strategic Industry Partnerships:** The unique "infrastructure" theme of the event will attract prominent automation, civil engineering, and technology sponsors, paving the way for exclusive internships, research grants, and direct placement opportunities for MACE students.
- **Establishment of a Space Robotics Hub:** Successfully organizing the inaugural AsRC formally launches MACE as the primary regional headquarters for future space-tech and robotics initiatives organized by RISA in South Asia.

10. Conclusion

The Asian Rover Challenge represents a bold, timely leap forward in robotics education. By focusing on the realistic future of space infrastructure, and by making this elite tier of engineering financially accessible to students across Asia, MACE has the opportunity to lead a global educational movement.

Hosting this challenge is more than organizing a competition—it is a long-term investment in global branding, industry alignment, and pioneering academic excellence. We confidently present this proposal to the MACE College Association to partner with us in shaping the future of planetary robotics.